

API SERVER ON ESP8266

Using MicroPython

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MicroPython is similar to Python 3, but it is meant for microcontrollers. MicroPython comes with an interactive Read-Evaluate-Print-Loop (REPL), which is an amazing feature that allows us to connect to a microcontroller, execute code quickly without the need to compile or upload code, and gives immediate feedback. Also, most importantly, the community support available for it makes the use of this firmware easier.

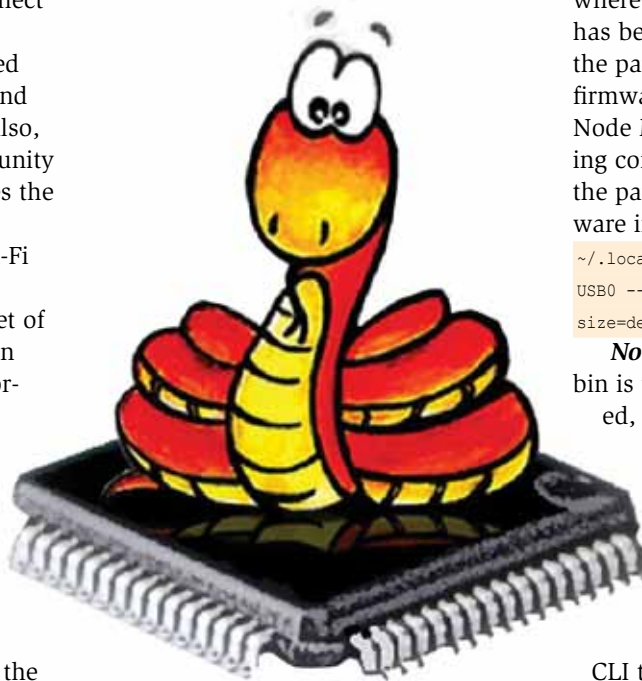
ESP8266 is a low-cost Wi-Fi module that is ideal for DIY projects involving the internet of things (IoT). As MicroPython is easy to use, which is important for beginners, it can be programmed easily. But the default firmware of ESP8266 and other ESP Wi-Fi boards are different. In this DIY project, we flash the MicroPython in Node MCU board and use the MicroPython to program the board and control its GPIO over network using Pico framework.

Flashing MicroPython firmware on the board

In order to use the PicoWeb framework and program the ESP in MicroPython to control GPIO on Node MCU, we need to erase the current firmware on ESP chip in the Node MCU. We can download the MicroPython firmware from its official website, which is in .bin format. The official website for downloading MicroPython firmware for the board is: <http://MicroPython.org/>

download/#esp8266

After downloading the firmware, we need to first erase the current firmware. For that, esptool is used to erase or flash new firmware into the ESP board. To install esptool, open



the command prompt and run the following command and install the esptool:

```
pip install esptool
```

Now, we need to connect Node MCU to the system and identify the port to which it is connected. If you are using a Linux based system, the command below will help you identify the port:

```
dmesg | grep tty
```

Next, we need to clean the board before flashing the MicroPython firmware with the below syntax. Please note, the port identified in

the above step is used in the syntax as -p.

```
~/local/bin/esptool.py -p /dev/ttyUSB0  
erase_flash
```

Now, we can flash the MicroPython firmware and open the folder where the MicroPython firmware has been downloaded. Then, copy the path and file name of that .bin firmware file and flash it to the Node MCU board using the following command. (You need to change the path and filename of the firmware in the command.)

```
~/local/bin/esptool.py --port /dev/tty  
USB0 --baud 460800 write_flash --flash_  
size=detect 0 esp8266-20210202-v1.14.bin
```

Note. esp8266-20210202-v1.14.bin is the firmware name downloaded, which needs to be replaced with the name of the firmware file in your system.

Wola! The board is ready, so we can enter the REPL prompt and explore. But to enter the REPL prompt we need another

CLI tool, named picocom, which can be installed through the command below:

```
sudo apt-get install picocom
```

Next, use the syntax below, which will display some information on the ESP board. Press **ctl + B** to enter the REPL prompt.

```
picocom /dev/ttyUSB0 -b115200
```

The command to soft reset the board from the REPL prompt is **ctl + D**

Pushing the file from your system onto the board will require cli tool adafruit-ampy. To install it in your system, use the following syntax: